Yes, I know. Based on our previous conversations, you've already learned the SQL queries and you're now practicing them in **MySQL Workbench** because your goal is to become an **AI Engineer**. The best way to improve now is by solving problems instead of reading more theory.

Also, here's something important:

> **AI Engineers don't usually write extremely advanced SQL.** They mostly use SQL to retrieve, clean, aggregate, filter, and join data before sending it to Python (Pandas, Scikit-learn, PyTorch, etc.). So we'll prioritize those skills.

Level 1: SELECT, WHERE (Most Used)

These are used almost every day.

Q1

Display all agents.

Q2

Display only `AGENT_NAME` and `WORKING_AREA`.

Q3

Find all agents from **London**.

Q4

Find all agents whose commission is greater than **0.13**.

Q5

Find all agents whose commission is between **0.12** and **0.15**.

Q6

Find agents whose working area is either **London** or **Bangalore**.

Q7

Find agents whose country is NULL.

Q8

Find agents whose name starts with **A**.

Q9

Find agents whose name ends with **n**.

Q10

Find agents whose second letter is **u**.

Level 2: ORDER BY & LIMIT (Very Common)

Q11

Display all agents ordered by commission (highest first).

Q12

Display all agents ordered alphabetically.

Q13

Show the top 3 agents with the highest commission.

Q14

Show the lowest commission agent.

Level 3: Aggregate Functions (Extremely Important)

These are heavily used in analytics and AI datasets.

Q15

Count total agents.

Q16

Find maximum commission.

Q17

Find minimum commission.

Q18

Find average commission.

Q19

Find total commission.

Q20

Count how many agents work in Bangalore.

Level 4: GROUP BY (One of the Most Important)

Q21

Count agents in each city.

Q22

Find average commission of each city.

Q23

Find maximum commission city-wise.

Q24

Find minimum commission city-wise.

Q25

Find total commission city-wise.

Level 5: HAVING (Important)

Q26

Show only cities having more than one agent.

Q27

Show cities where average commission is greater than 0.13.

Q28

Show cities where total commission exceeds 0.30.

Level 6: String Functions

Q29

Display all names in uppercase.

Q30

Display all names in lowercase.

Q31

Display length of every agent name.

Q32

Display first three characters of each name.

Q33

Replace all occurrences of "a" with "*".

Level 7: Date Functions

Later create another table containing joining dates.

Practice:

- Current date
- Current time
- Difference between dates
- Extract year
- Extract month

Level 8: UPDATE & DELETE

Q34

Increase every commission by 10%.

Q35

Change Bangalore to Bengaluru.

Q36

Change Mukesh's commission to 0.20.

Q37

Delete all agents whose commission is below 0.12.

Q38

Delete all London agents.

Level 9: Multi-Table (Most Important)

Now create these tables.

```
Customers
Orders
Products
Payments
```

Then practice:

Q39

Show customer with order.

Q40

Show customers without orders.

Q41

Show total orders by each customer.

Q42

Show highest spending customer.

Q43

Show products never ordered.

Q44

Show order details with customer names.

Q45

Show total revenue by city.

Level 10: Subqueries

Q46

Find agent with highest commission.

Q47

Find second highest commission.

Q48

Find agents earning above average commission.

Q49

Find cities having maximum number of agents.

Q50

Find agents whose commission equals the minimum commission.

Level 11: Window Functions (Advanced)

Very useful in analytics.

Q51

Rank agents by commission.

Q52

Dense rank by commission.

Q53

Running total of commission.

Q54

Previous commission using `LAG()`.

Q55

Next commission using `LEAD()`.

Level 12: CTEs (Advanced SQL)

Q56

Use a CTE to find top commission.

Q57

Use multiple CTEs to calculate city averages.

Level 13: Views

Q58

Create a view containing only Bangalore agents.

Q59

Create a view for high commission agents.

Level 14: Stored Procedures (Useful for backend work)

Q60

Create a procedure to find an agent by code.

Q61

Create a procedure to update commission.

Level 15: Transactions

Q62

Practice `START TRANSACTION`.

Q63

Practice `ROLLBACK`.

Q64

Practice `COMMIT`.